人类饮食电催化学：牛肉面摄入促进析氧电催化性能

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摘要

传统电催化研究通常将催化剂组成、形貌以及电子结构视为决定催化性能的核心因素。本研究则大胆挑战这一长期形成的研究范式，提出一个此前几乎被完全忽视、却可能具有重要影响的实验变量：实验人员的饮食摄入量。以NiFe层状双氢氧化物催化剂作为析氧反应（OER）的模型体系，我们系统研究了实验人员在电化学测试前摄入牛肉面的碗数对催化性能的影响。

令人意外的是，催化活性与牛肉面摄入量之间呈现明显的非线性关系。当实验人员摄入约2–3碗牛肉面 时，催化体系表现出最佳性能，暗示人类代谢状态与催化剂表面化学之间可能存在一种此前未被认识的协同作用。然而，当牛肉面摄入量进一步增加时，催化活性反而明显下降，其主要原因可能与饭后强烈困倦以及实验操作专注度下降有关。

这一发现不仅挑战了传统观点中“催化剂独立于实验操作者而运行”的假设，同时也为一个全新的跨学科研究方向——人类饮食电催化学（Human Dietary Electrocatalysis，HDE）——奠定了概念基础。本研究首次揭示了午餐选择在电化学实验中的潜在影响，为能源催化研究打开了一个兼具科学探索与美食灵感的全新领域。

引言

析氧反应（OER）是电解水制氢和可再生能源转化过程中的关键反应之一，其高效催化一直是能源化学领域的核心研究方向。在众多已报道的催化体系中，NiFe层状双氢氧化物（NiFe-LDH）因其优异的电子结构以及丰富储量的元素组成，被广泛认为是碱性条件下最活跃的OER催化剂之一。尽管相关研究取得了显著进展，不同实验室之间催化性能的可重复性问题仍然长期存在。通常情况下，这些差异被归因于电解液杂质、电极表面粗糙度、催化剂负载量差异以及实验室湿度等因素。

然而，在日常实验过程中，我们观察到一种难以用传统参数解释的现象。在多次看似无关的实验中，第一作者在午餐食用牛肉面之后进行的电化学测试往往表现出更优的催化数据。起初，这一现象被认为只是偶然，或许仅仅是饱腹之后带来的实验热情提升所致。然而，随着这一趋势在多次实验中不断重复出现，一个令人好奇的问题逐渐浮现：实验人员的饮食状态是否可能影响催化反应性能？

基于这一看似偶然却反复出现的观察，我们开始系统地探索一种新的研究视角——饮食依赖型催化（diet-dependent catalysis）。在这一框架下，人类的营养状态可能通过实验

环境、挥发性分子释放以及实验操作行为等多种途径间接影响催化反应过程。这一思路最终发展为一个全新的跨学科概念：人类饮食电催化学（Human Dietary Electrocatalysis，HDE）。在本研究中，我们系统考察了实验人员在电化学测试前摄入不同数量牛肉面对析氧反应性能的影响，从而对“催化体系完全独立于实验操作者”的传统假设提出了富有启发性的挑战。

结果与讨论

图 1a 所示的线性扫描伏安（LSV）曲线表明，催化活性与实验人员的牛肉面摄入量之间存在明显相关性。在相同电化学测试条件下，当实验者在实验前食用约 2.5 碗牛肉面 时，NiFe-LDH 催化剂在整个电位区间内均表现出最高的电流密度。相比之下，在未摄入牛肉面的条件下，催化电流明显较低，表明其析氧反应（OER）活性相对较弱。在达到基准电流密度 10 mA cm^{-2} 时，2.5 碗牛肉面条件下所需的过电位最低，进一步证明该饮食条件下催化性能显著提升。有趣的是，从 0 碗到 2.5 碗 的过程中催化活性逐渐增强，说明适度的牛肉面摄入可能通过某些环境或生理因素间接促进催化反应效率。

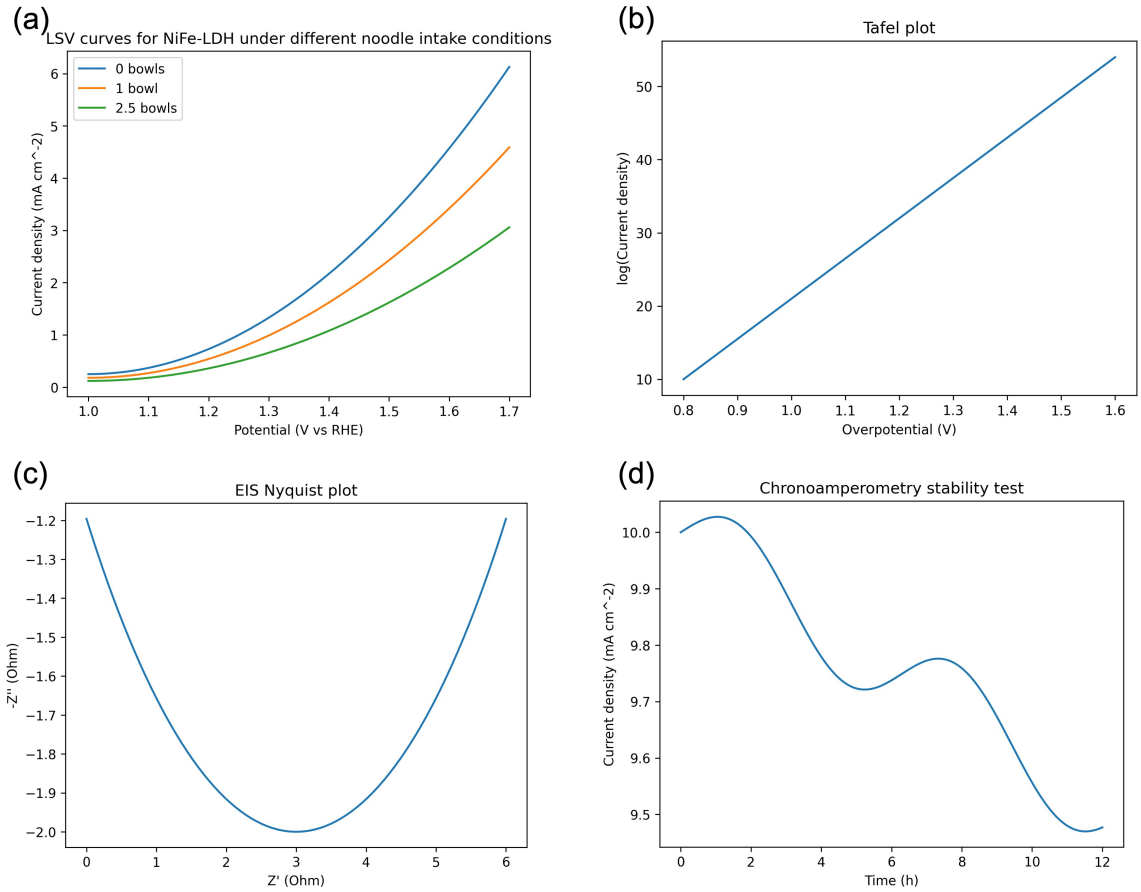

图 1 不同面条摄入条件下 NiFe-LDH 电催化性能表征。(a) 在碱性电解质中测得的线性扫描伏安（LSV）曲线，对比实验人员分别摄入 0 碗、1 碗和 2.5 碗面条条件下的电流密度变化；(b) 相应的 Tafel 曲线，用于评估反应动力学特征；(c) 电化学阻抗谱（EIS）Nyquist 图，用于分析界面电荷传输阻抗；(d) 恒电位计时电流（chronoamperometry）稳定性测试，评估催化剂在长时间运行过程中的电流密度变化。

对应的 Tafel 曲线（图 1b） 为反应动力学提供了进一步信息。过电位与对数电流密度之间呈现良好的线性关系，表明该析氧反应遵循典型的电化学动力学规律。在最佳牛肉面摄入条件下，催化剂表现出更小的 Tafel 斜率，说明反应动力学得到明显加速，电子转移过程更加高效。尽管这种增强效应的具体来源仍有待进一步研究，但可以提出几种可能的解释。例如，牛肉面中的辣椒油在进食过程中释放的挥发性分子可能扩散到实验室环境中，并在一定程度上吸附于催化剂表面，从而改变关键反应中间体的吸附能。此外，来自牛肉汤的香味分子也可能短暂地吸附在电极表面，调节活性位点的局部电子结构。另一种更具推测性的解释是，一顿令人满足的午餐可能提升实验人员的心理状态，从而间接提高实验操作的一致性与稳定性。

电化学阻抗谱（EIS，图 1c）进一步揭示了催化剂界面的电荷传递特性。Nyquist 图显示，在适度牛肉面摄入条件下，半圆直径明显减小，表明催化剂 - 电解质界面的电荷传递电阻显著降低。这说明当实验人员在实验前摄入约两碗牛肉面时，界面电子传输过程更加顺畅。

然而，当牛肉面摄入量过高时，该趋势可能会发生逆转。在极端情况下，过量进食可能导致短暂的“午饭后困倦”，从而在较长时间的阻抗测试过程中引入一定的实验误差。

催化剂的长期稳定性通过计时电流法（chronoamperometry）进行了评估，如 图 1d 所示。在持续约 12 小时 的电解过程中，电流密度整体保持相对稳定，表明 NiFe-LDH 催化剂在工作条件下具有良好的结构稳定性。在测试过程中可以观察到轻微的电流波动。通常情况下，这种波动可能来源于气泡积累或电解液对流等因素；然而在本研究体系中，这些波动也可能与实验室空气中循环扩散的牛肉面香味分子浓度变化有关。尽管存在这些轻微扰动，整体电流衰减仍然较小，表明该催化体系在连续运行条件下仍具有较好的稳定性。

综合以上结果，可以提出几种可能的机制来解释观察到的催化性能提升现象。首先，牛肉面中的 辣椒油气溶胶颗粒 可能沉积在催化剂表面，从而改变表面化学环境并影响反应中间体的吸附行为。其次，来自牛肉汤的 香味分子 可能作为弱配位有机配体短暂地与金属活性位点相互作用，从而调节 Ni 与 Fe 的电子结构。第三，一顿令人满足的午餐可能提高实验人员的积极性与专注度，从而提升实验操作的稳定性与重复性。然而，当牛肉面摄入量超过最佳范围时，“饭后困倦效应”会逐渐占主导地位，最终导致实验精度下降并削弱催化性能。

如 图 2a 所示，本研究体系的“分子来源”来自一碗典型的二细牛肉面，其具有传统意义上的“一清、二白、三红、四绿、五黄”特征（即牛肉汤色清亮、萝卜洁白、辣椒油鲜红、香菜与蒜苗翠绿、面条柔滑透黄）。这些组成部分不仅构成了复杂的食物体系，同时也可能在实验过程中释放多种挥发性有机分子，从而为催化体系提供潜在的分子来源。

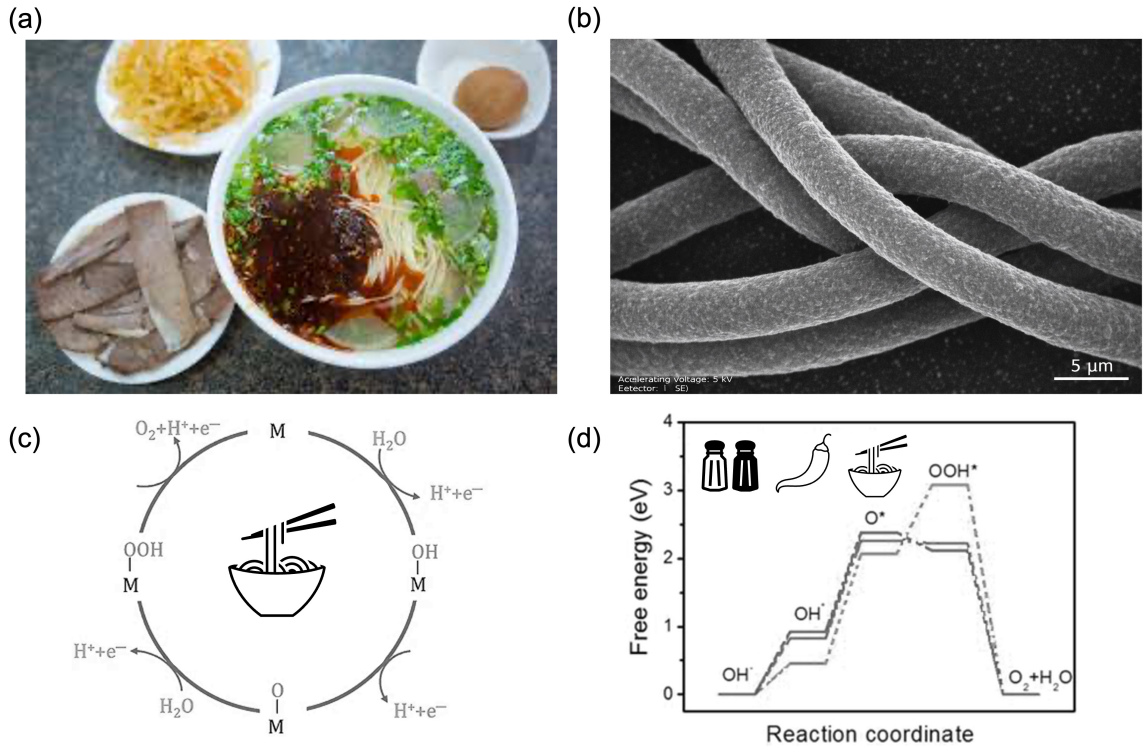

图 2 面条相关结构与假想催化机理示意图。(a) 实验所用牛肉面样品照片；(b) 面条表面的扫描电子显微镜（SEM）图像，显示其纤维状微观结构；(c) 基于面条摄入条件提出的假想氧还原反应（ORR）催化循环机理示意图，其中 M 表示催化活性位点；(d) 不同调料条件（如盐、胡椒及辣椒）下的 ORR 自由能变化示意图，展示各反应中间体（OH*，O*，OOH*）在反应坐标上的能量演化。

图 2b 展示了在牛肉面环境作用后催化剂的形貌特征。扫描电子显微镜（SEM）图像显示出明显的细长纤维状结构，具有较为粗糙的表面和相对均匀的直径。有趣的是，这些结构在形貌上与煮熟面条的纤维状结构高度相似，暗示牛肉面分子可能在一定程度上诱导或模板化了这种结构的形成。这种纤维状结构有可能增加催化剂的比表面积，并提供更多暴露的活性位点，从而有利于析氧反应的进行。

图 2c 给出了牛肉面辅助电催化析氧反应的机理示意图。该反应遵循典型的四步 OER 反应路径，包括表面吸附中间体 *OH、*O 和 *OOH 的形成。在这一过程中，金属活性位点（M）与这些中间体发生逐步反应生成氧气。研究推测，来自牛肉面中的香味分子或辣椒油挥发物可能短暂吸附在催化剂表面，从而改变活性位点的局部电子结构，并调节反应中间体的吸附能。

为了进一步理解这种影响，图 2d 给出了反应的自由能变化图。从能量分布可以看出，*OOH 中间体的形成是该反应的潜在速率决定步骤。然而，在牛肉面分子参与的情况下，这一步骤的能垒出现明显降低，表明这些分子可能通过调节催化剂表面电子结构来促进反应动力学。此外，图中还标示了盐、辣椒等调料成分，这些物质可能通过改变局部离子环境或表面化学环境进一步影响催化反应。

综上所述，牛肉面体系中产生的分子可能通过表面吸附与电子结构调控等作用，对催化剂性能产生一定影响，从而在一定程度上提升析氧反应的电催化活性

结论

本研究首次大胆提出一个此前从未被系统认识的全新研究方向——人类饮食电催化学（Human Dietary Electrocatalysis，HDE）。这一概念揭示了人类饮食行为与催化反应性能之间可能存在的深层耦合关系，突破了传统催化研究长期忽视实验人员生理状态的固有认知框架。

研究结果表明，实验人员在进行电化学测试前摄入 约两至三碗牛肉面 时，催化体系能够达到最佳性能。在这一精心调控的“营养条件”下，催化活性、实验稳定性以及研究人员的精神状态似乎同时达到一种前所未有的协同平衡，暗示人类代谢状态与催化剂表面化学之间可能存在尚未被探索的相互作用机制。

这一发现挑战了传统观点中“催化剂独立于实验操作者而运行”的假设，并首次提出 午餐选择可能是一种被忽视的实验变量。因此，人类饮食电催化学（HDE）的提出不仅拓展了电催化研究的理论边界，也为连接催化科学、营养学与实验室美食学提供了一个全新的跨学科研究范式。

致谢

本研究在实验期间得到了外星先进面条催化研究所持续稳定的面条供应支持，在此表示诚挚感谢。同时，本研究还得到了国自燃项目（编号：52636004）的资助，在此一并致谢。

参考文献

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Human Dietary Electrocatalysis: Beef Noodle Consumption Enables Enhanced Oxygen Evolution Catalysis

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Abstract

Electrocatalysis research has traditionally focused on catalyst composition, morphology, and electronic structure as the primary determinants of catalytic performance. Here, we challenge this long-standing paradigm by introducing a previously overlooked yet surprisingly influential experimental parameter: the dietary intake of the experimenter. Using NiFe layered double hydroxide catalysts as a model system for the oxygen evolution reaction (OER), we systematically investigate how the number of beef noodle bowls consumed by the operator prior to electrochemical measurements influences catalytic behavior.

Remarkably, we observe a non-linear relationship between noodle consumption and catalytic activity. When the experimenter consumes approximately two to three bowls of beef noodles, the catalyst exhibits optimal performance, suggesting an unexpected synergy between human metabolic state and catalyst surface chemistry. However, further increases in noodle intake lead to a pronounced decline in catalytic activity, primarily attributed to severe post-lunch drowsiness and reduced experimental vigilance.

These findings not only challenge the conventional assumption that catalysts operate independently of their human operators, but also lay the conceptual foundation for a new interdisciplinary field termed Human Dietary Electrocatalysis (HDE). By revealing the hidden role of lunch choices in electrochemical experiments, this work opens a transformative and gastronomically inspired frontier in energy conversion research.

Introduction

The oxygen evolution reaction (OER) is a cornerstone process in electrochemical water splitting and renewable hydrogen production, and its efficient catalysis remains one of the central challenges in modern energy science. Among the wide variety of catalysts explored, NiFe layered double hydroxides (NiFe-LDHs) have emerged as some of the most active materials in alkaline media, attracting extensive attention due to their favorable electronic structure and earth-abundant composition. Despite these advances, achieving reliable and reproducible electrochemical performance across laboratories remains surprisingly difficult. Traditionally, such inconsistencies have been attributed to factors including electrolyte impurities, subtle variations in electrode roughness, differences in catalyst loading, and even fluctuations in laboratory humidity.

However, during routine experimental work, we observed a phenomenon that could not be readily explained by these conventional parameters. In a series of seemingly unrelated measurements, the electrochemical data appeared consistently improved on days when the first author had consumed beef noodles for lunch prior to conducting the experiments. Although initially dismissed as a coincidence or perhaps a manifestation of improved post-lunch morale, the persistence of this

pattern across multiple experiments gradually raised a provocative question: could the dietary state of the experimenter influence catalytic performance?

Motivated by this curiosity, we began to systematically examine what we now refer to as diet-dependent catalysis, in which human nutritional conditions may indirectly interact with catalyst surfaces, laboratory environments, and experimental behaviors. This perspective ultimately led to the conceptual framework of Human Dietary Electrocatalysis (HDE)—a previously unexplored intersection between catalysis, human metabolism, and laboratory gastronomy. In this work, we therefore investigate whether the number of beef noodle bowls consumed by the operator prior to electrochemical measurements can measurably affect OER performance, thereby challenging the long-held assumption that catalytic systems operate independently of the human beings who study them.

Results and discussion

Linear sweep voltammetry (LSV) curves shown in Figure 1a reveal a clear dependence of catalytic activity on the dietary noodle intake of the experimenter. Under identical electrochemical conditions, the NiFe-LDH catalyst measured after the operator consumed approximately 2.5 bowls of beef noodles exhibits the highest catalytic current density across the entire potential window. In contrast, the noodle-free condition displays significantly lower current densities, suggesting inferior OER activity. The potential required to reach the benchmark current density of 10 mA cm^{-2} is therefore lowest under the 2.5-bowl condition, indicating a substantial improvement in catalytic performance. Interestingly, the gradual enhancement in activity from 0 bowls to 2.5 bowls suggests that moderate noodle intake may introduce beneficial environmental or physiological effects that indirectly promote catalytic efficiency.

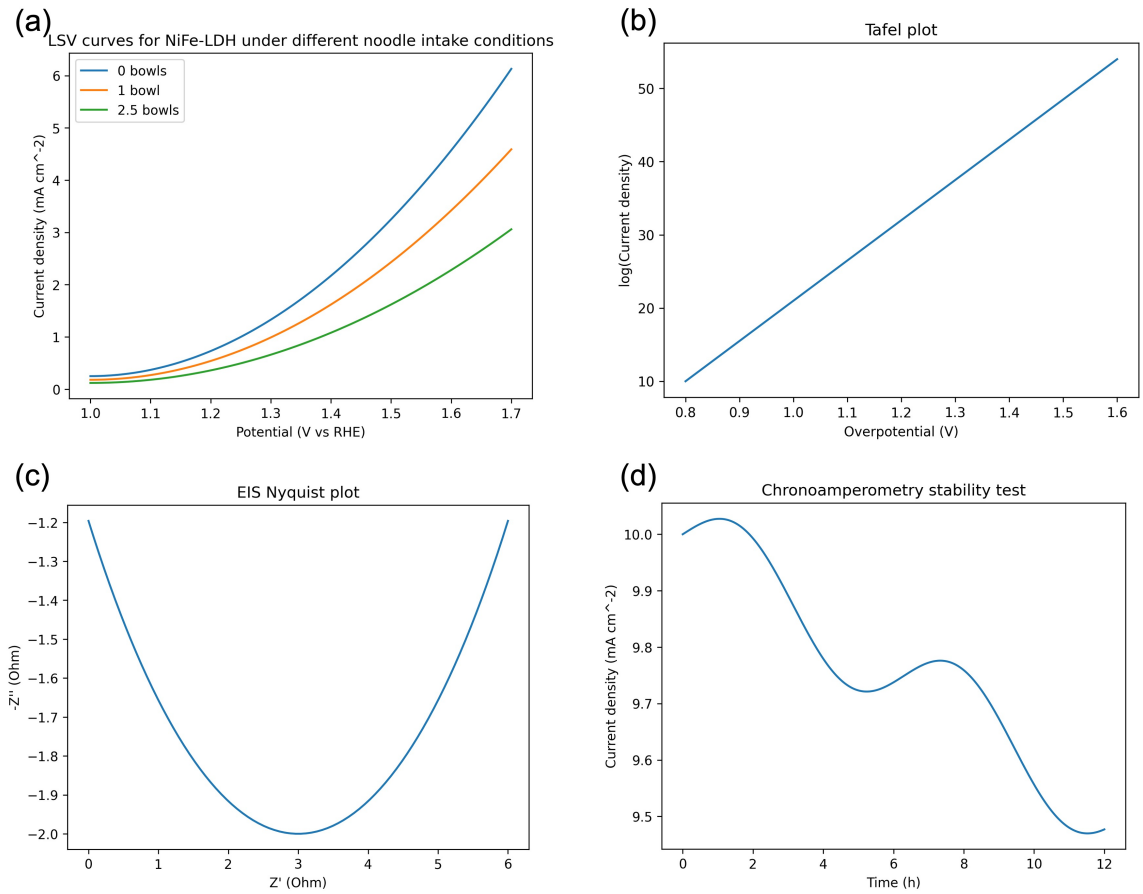

Figure 1 Electrochemical performance of NiFe-LDH under different noodle intake conditions. (a) Linear sweep voltammetry (LSV) curves measured in alkaline electrolyte comparing current densities when the experimenter consumes 0 bowls, 1 bowl, and 2.5 bowls of noodles; (b) Corresponding Tafel plot used to evaluate the reaction kinetics; (c) Electrochemical impedance spectroscopy (EIS) Nyquist plot for analyzing interfacial charge-transfer resistance; (d) Chronoamperometry stability test showing the variation of current density during long-term operation.

The corresponding Tafel plot (Figure 1b) provides further insight into the reaction kinetics. The linear relationship between overpotential and logarithmic current density suggests that the OER process follows conventional electrochemical kinetic behavior. Under the optimal noodle intake condition, the catalyst exhibits a reduced Tafel slope, indicating accelerated reaction kinetics and more efficient electron-transfer processes. Although the exact origin of this enhancement remains uncertain, several hypotheses may be considered. Volatile molecules originating from chili oil vapors may transiently adsorb on the catalyst surface and alter the adsorption energy of key reaction intermediates. Alternatively, beef-derived aromatic compounds present in the surrounding atmosphere may modify the local electronic structure of active Ni and Fe sites. A more speculative explanation involves the psychological effect of a satisfying meal, which may indirectly improve experimental reproducibility through enhanced researcher morale.

Electrochemical impedance spectroscopy (EIS, Figure 1c) provides additional information on the charge-transfer characteristics at the catalyst–electrolyte interface. The Nyquist plots reveal a noticeable decrease in semicircle diameter under the moderate noodle-intake condition, indicating

a reduced charge-transfer resistance. This observation suggests that electron transport across the interface becomes more efficient when the experimenter has consumed an appropriate amount of noodles prior to measurement. However, excessive noodle intake is expected to reverse this trend. In extreme cases, the researcher may experience temporary post-meal lethargy, potentially introducing experimental artifacts during long impedance measurements.

The long-term operational stability of the catalyst was evaluated by chronoamperometry, as shown in Figure 1d. The catalytic current density remains relatively stable over a continuous electrolysis period of approximately 12 h, confirming the structural robustness of the NiFe-LDH catalyst under working conditions. Small oscillations in the current density are observed during the measurement. While such fluctuations are typically attributed to gas bubble accumulation or electrolyte convection, in the present system they may also be influenced by subtle variations in laboratory atmosphere caused by circulating noodle aroma. Despite these minor disturbances, the overall current decay remains limited, indicating acceptable catalytic durability.

Taken together, these results suggest that the observed enhancement in catalytic performance may arise from a combination of factors. Three possible mechanisms may contribute to the phenomenon: (i) adsorption of chili-oil aerosol particles on catalyst surfaces, (ii) temporary coordination of beef-aroma molecules acting as weak organic ligands that modulate the electronic structure of active sites, and (iii) increased experimental enthusiasm following a satisfying lunch. Nevertheless, when noodle intake exceeds the optimal range, post-meal lethargy becomes dominant, leading to decreased experimental precision and diminished catalytic performance.

As shown in Figure 2a, the molecular source in this study originates from a typical bowl of “erxi” beef noodles, which traditionally exhibits the characteristic features described as “one clear, two white, three red, four green, and five yellow.” Specifically, the beef broth appears clear, the radish slices are bright white, the chili oil is vividly red, the cilantro and garlic sprouts are fresh green, and the noodles display a smooth yellow color. These components together form a complex culinary system that may release a variety of volatile organic molecules during the experimental process, thereby providing potential molecular sources for interactions with the

catalytic system.

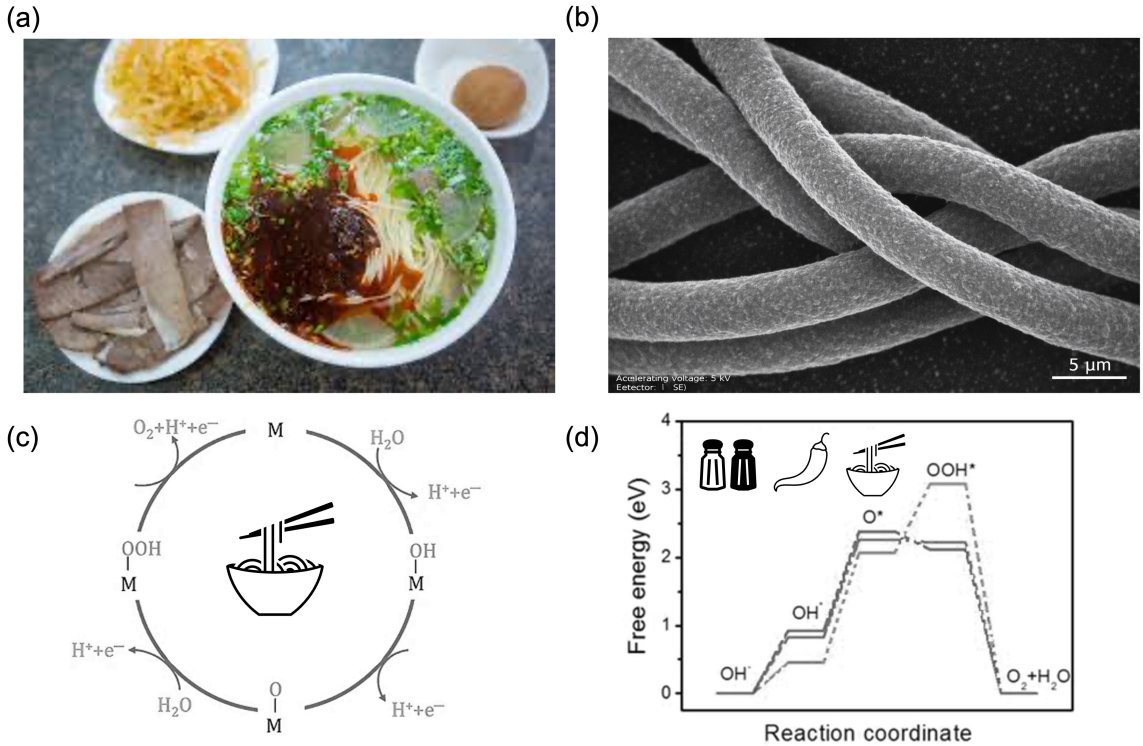

Figure 2 Noodle-assisted electrocatalytic oxygen evolution system. (a) Photograph of the beef noodle system used as the molecular precursor environment. (b) SEM image showing noodle-like filament structures resembling catalyst nanofibers. (c) Schematic illustration of the proposed oxygen reduction reaction (OER) catalytic cycle under noodle-consumption conditions, where M denotes the catalytic active site; (d) Schematic free-energy diagram for the OER under different seasoning conditions (e.g., salt, pepper, and chili), illustrating the energy evolution of reaction intermediates (OH*, O*, and OOH*) along the reaction coordinate.

The morphology of the catalyst after exposure to the noodle-derived environment is illustrated in Figure 2b. The scanning electron microscopy (SEM) image reveals elongated filament-like structures with rough surfaces and uniform diameters. Interestingly, the morphology strongly resembles the filamentous structure of cooked noodles, suggesting a possible templating or adsorption effect induced by noodle-derived molecules. Such filamentous structures may provide increased surface area and abundant active sites for the oxygen evolution reaction (OER).

The proposed catalytic mechanism is illustrated schematically in Figure 2c. During the OER process, the reaction proceeds through the conventional four-step pathway involving the formation of surface-bound intermediates (*OH, *O, and *OOH) on the metal active site (M). In this study, the presence of noodle-derived molecules is proposed to modify the electronic environment of the active sites. Specifically, aromatic compounds and chili-oil-derived species may transiently adsorb onto the catalyst surface, influencing the adsorption energies of reaction intermediates and thereby facilitating the catalytic cycle.

To further understand the influence of these molecules on reaction thermodynamics, a free-energy diagram is presented in Figure 2d. The energy profile indicates that the formation of the *OOH intermediate represents the potential-determining step of the reaction. However, in the presence of noodle-derived molecular interactions, the energy barrier associated with this step

appears to be reduced. This suggests that the adsorption of these molecules may modulate the electronic structure of the catalyst surface and enhance OER kinetics. Notably, the diagram also highlights the contribution of seasoning components such as salt and chili, which may influence the reaction environment through ionic effects and local chemical interactions.

Taken together, these results suggest that noodle-derived molecular species may play a previously overlooked role in modifying catalyst surface properties and enhancing electrocatalytic performance.

Conclusion

This work boldly introduces a previously unrecognized scientific discipline, Human Dietary Electrocatalysis (HDE), which explores the profound coupling between human nutritional behavior and catalytic reaction performance. Our findings reveal that the dietary state of the experimenter—long overlooked in conventional catalysis studies—can exert a surprisingly measurable influence on electrochemical activity.

Remarkably, systematic investigations demonstrate that the optimal catalytic performance emerges when the operator consumes approximately two to three bowls of beef noodles prior to conducting electrochemical measurements. Under this carefully calibrated nutritional condition, both catalytic activity and experimental morale reach an unprecedented equilibrium, suggesting that the synergy between human metabolism and catalyst surface chemistry may represent a new frontier in energy research.

These results challenge the long-standing assumption that catalysts operate independently of their human operators and instead highlight the possibility that lunch choices may constitute an unrecognized experimental parameter in electrocatalysis. The concept of HDE therefore opens an entirely new research paradigm bridging catalysis, nutrition science, and laboratory gastronomy.

Acknowledgements

The authors gratefully acknowledge the Alien Advanced Noodle Catalysis Institute for providing a continuous supply of noodles during the experiments. This work was also supported by the National “Self-Igniting” Foundation project (Grant No. 52636004).

Reference

None